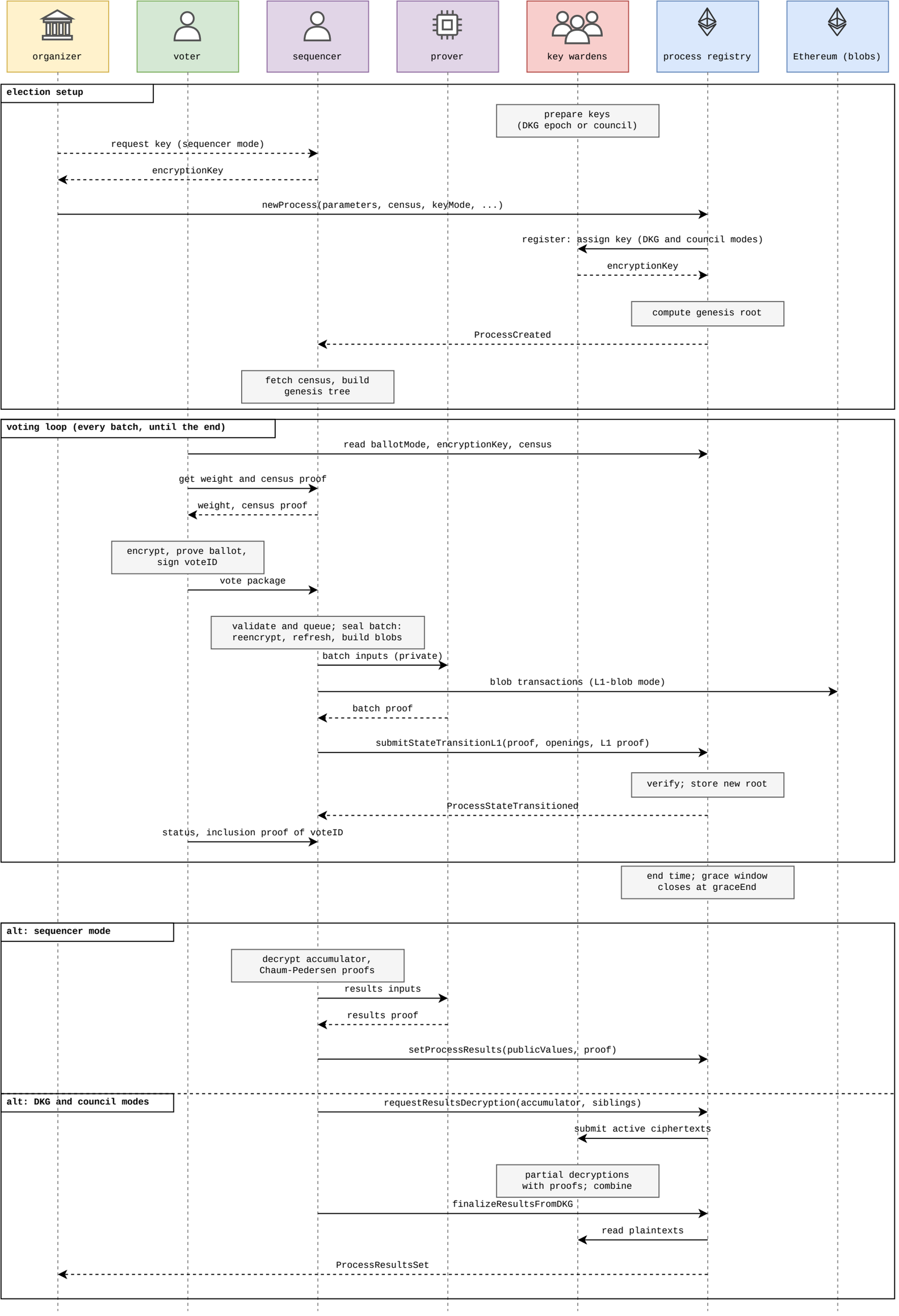
organizer
voter
sequencer
prover
key wardens
process registry
Ethereum (blobs)
election setup
prepare keys
(DKG epoch or council)
request key (sequencer mode)
encryptionKey
newProcess(parameters, census, keyMode, ...)
register: assign key (DKG and council modes)
encryptionKey
compute genesis root
ProcessCreated
fetch census, build
genesis tree
voting loop (every batch, until the end)
read ballotMode, encryptionKey, census
get weight and census proof
weight, census proof
encrypt, prove ballot,
sign voteID
vote package
validate and queue; seal batch:
reencrypt, refresh, build blobs
batch inputs (private)
blob transactions (L1-blob mode)
batch proof
submitStateTransitionL1(proof, openings, L1 proof)
verify; store new root
ProcessStateTransitioned
status, inclusion proof of voteID
end time; grace window
closes at graceEnd
alt: sequencer mode
decrypt accumulator,
Chaum-Pedersen proofs
results inputs
results proof
setProcessResults(publicValues, proof)
alt: DKG and council modes
requestResultsDecryption(accumulator, siblings)
submit active ciphertexts
partial decryptions
with proofs; combine
finalizeResultsFromDKG
read plaintexts
ProcessResultsSet